

Deep Generative Learning: Denoising Diffusion Model

Deep Generative Learning

learning to generative data:

Given observed samples x from a distribution of interest, the goal of a generative model is to learn to model its true data distribution $p(x)$.

samples data from true data distribution, train neural network with data, samples from neural network to create generative distribution

application

content generation

text to image generation

representation learning

artistic tools

image super-resolution

denosing diffusion probabilistic models

2 process:

forward diffusion process: add noise

gradually add noise to input

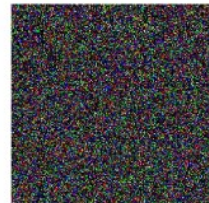
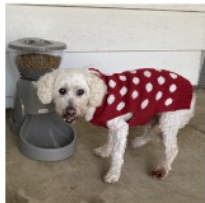
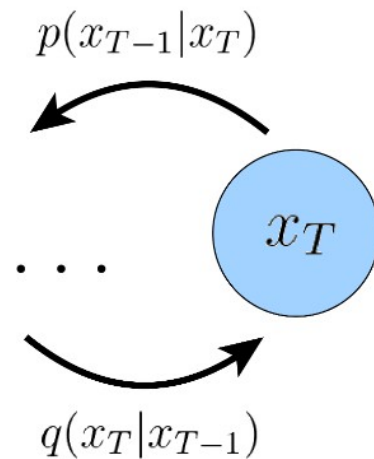
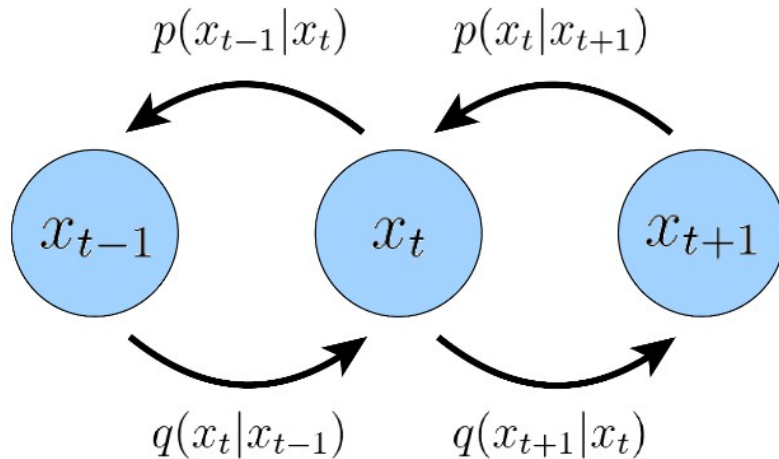
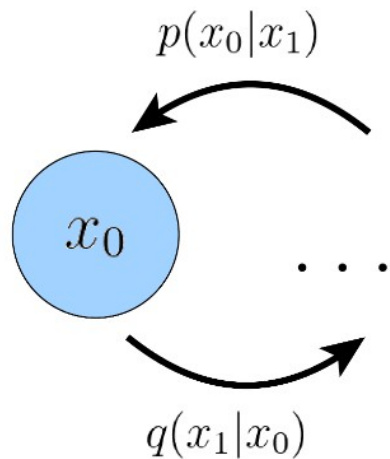
reverse diffusion process: denoise

learns to generative data

Related Work

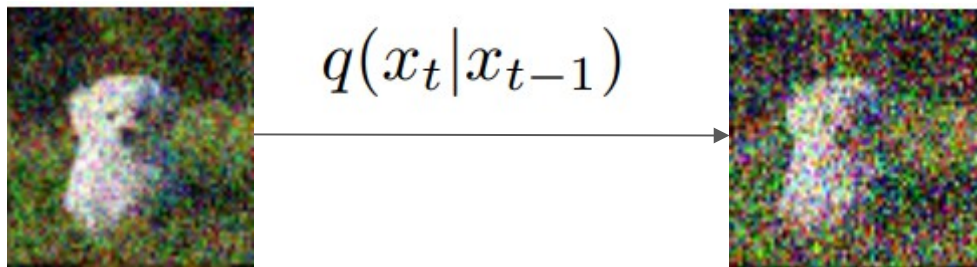
Math part

Notation



Forward process

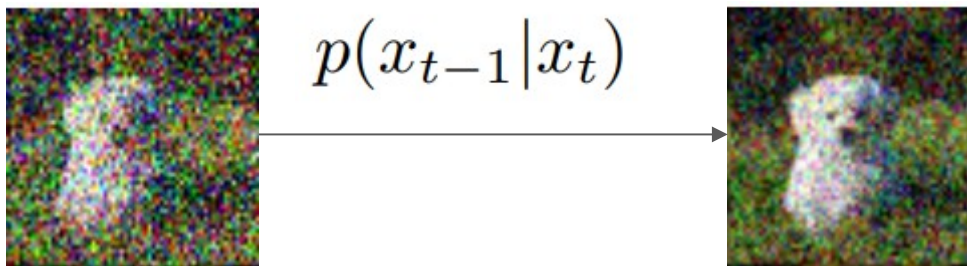
$q(x_t|x_{t-1})$: object distribution that given by a less noised object



$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

Reverse process

$p(x_{t-1} | x_t)$: object distribution that given by a more noised object



$$p_{\theta}(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t))$$

Maximum likelihood estimation(MLE)

$$\max_{\theta} \log \Pr(x_0)$$

$$\log p(x_0) = \log \int p(x_0, x_{1:T}) dx_{1:T}$$

Evidence Lower Bound (ELBO)

$$\begin{aligned}\log p(x_0) &= \log \int p(x_0, x_{1:T}) dx_{1:T} \\ &= \log \int \frac{p(x_0, x_{1:T}) q(x_{1:T} | x_0)}{q(x_{1:T} | x_0)} dx_{1:T} \\ &= \log \mathbb{E}_{q(x_{1:T} | x_0)} \left[\frac{p(x_0, x_{1:T})}{q(x_{1:T} | x_0)} \right] \\ &\leq \mathbb{E}_{q(x_{1:T} | x_0)} \left[\log \frac{p(x_0, x_{1:T})}{q(x_{1:T} | x_0)} \right]\end{aligned}$$

Evidence Lower Bound (ELBO)

$$\begin{aligned} ELBO &= \mathbb{E}_{q(x_{1:T}|x_0)} \left[\log \frac{p(x_0, x_{1:T})}{q(x_{1:T}|x_0)} \right] \\ &= L_1 - \sum_{t=2}^T L_t - L_{T+1} \\ L_1 &:= \log p_\theta(x_0|x_1) \\ L_t &:= D_{KL}(q(x_{t-1}|x_t, x_0) || p_\theta(x_{t-1}|x_t)) \\ L_{T+1} &:= D_{KL}(q(x_T|x_0) || p(x_T)) \end{aligned} \tag{21}$$

1. Reconstruction term(L_1): predicting the log probability of the original data sample given the first-step latent.

2. Denoising matching term(L_t): We use a neural network to learn the reverse diffusion process $p_\theta(x_{t-1}|x_t)$ to approximate the ground-truth denoising process $q(x_{t-1}|x_t, x_0)$. Therefore, this term is minimized when the two distributions match as closely as possible, as measured by their KL Divergence.

3. Prior matching term(L_{T+1}): this term represents how close the distribution of the final noisified input is to the standard Gaussian prior. It is equal to zero(two distributions are the same) if we learn a perfect model.

$$L_1 := \log p_{\theta}(x_0|x_1)$$

Reconstruction term:

Predicting the log probability of the original data sample given the first-step noised image, it can be approximated and optimized using a Monte Carlo estimate.

$$L_{T+1} := D_{KL}(q(x_T|x_0)||p(x_T))$$

Prior matching term:

Represents how close the distribution of the final noisified input is to the standard Gaussian prior. It is equal to zero (two distributions are the same) if we learn a perfect model.

$$\alpha_t = 1 - \beta_t$$

Recall the forward process

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$$

Sample object

reparam $x \sim \mathcal{N}(x; \mu, \sigma^2)$
 $x = \mu + \sigma\epsilon, \quad \epsilon \sim \mathcal{N}(0, 1)$

$$\begin{aligned}x_t &= \sqrt{1 - \beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon_{t-1} \\&= \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_{t-1} \\&= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\alpha_t(1 - \alpha_{t-1})}\epsilon_{t-2} + \sqrt{1 - \alpha_t}\epsilon_{t-1} \\&= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\sqrt{\alpha_t - \alpha_t\alpha_{t-1}}^2 + \sqrt{1 - \alpha_t}^2}\epsilon_{t-2} \\&= \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{1 - \alpha_t\alpha_{t-1}}\epsilon_{t-2} \\&\dots\dots \\&= \sqrt{\alpha_t\alpha_{t-1}\dots\alpha_0}x_0 + \sqrt{1 - \alpha_t\alpha_{t-1}\dots\alpha_0}\epsilon_0 \\&= \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_0\end{aligned}$$

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

Recall the forward process

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$$

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$$

$$L_t := D_{KL}(q(x_{t-1}|x_t, x_0) || p_{\theta}(x_{t-1}|x_t))$$

Denoising matching term:

Use a neural network to learn the reverse diffusion process $p_{\theta}(x_{t-1}|x_t)$ to approximate the ground-truth denoising process $q(x_{t-1}|x_t, x_0)$. Therefore, this term is minimized when the two distributions match as closely as possible, as measured by their KL Divergence.

$$\begin{aligned}
q(x_{t-1}|x_t, x_0) &= \frac{q(x_t|x_{t-1}, x_0)q(x_{t-1}|x_0)}{q(x_t|x_0)} \\
&= \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)} \\
&= \frac{\mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)\mathcal{N}(x_t; \sqrt{\bar{\alpha}_{t-1}}x_0, (1 - \bar{\alpha}_{t-1})I)}{\mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)} \\
&\propto \mathcal{N}(x_{t-1}; \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)x_0}{1 - \bar{\alpha}_t}, \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}I)
\end{aligned}$$

Loss function

$$\begin{aligned} & \arg \min_{\theta} D_{KL}(q(x_{t-1}|x^t, x_0) || p_{\theta}(x_{t-1}|x_t)) \\ &= \arg \min_{\theta} D_{KL}(\mathcal{N}(x_{t-1}; \mu_q, \Sigma) || \mathcal{N}(x_{t-1}; \mu_{\theta}, \Sigma)) \\ &= \arg \min_{\theta} \frac{1}{2\sigma^2} \left[||\mu_{\theta} - \mu_q||_2^2 \right] \end{aligned}$$

Simplify the Loss function

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_0$$

$$x_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_0}{\sqrt{\bar{\alpha}_t}}$$

$$\begin{aligned}\mu_q &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)x_0}{1 - \bar{\alpha}_t} \\ &= \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)}\epsilon_0\end{aligned}$$

$$\mu_\theta = \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)}\epsilon_\theta(x_t, t)$$

Loss function

$$\begin{aligned} & \arg \min_{\theta} D_{KL}(q(x_{t-1}|x^t, x_0) || p_{\theta}(x_{t-1}|x_t)) \\ &= \arg \min_{\theta} D_{KL}(\mathcal{N}(x_{t-1}; \mu_q, \Sigma) || \mathcal{N}(x_{t-1}; \mu_{\theta}, \Sigma)) \\ &= \arg \min_{\theta} \frac{1}{2\sigma^2} \left[||\mu_{\theta} - \mu_q||_2^2 \right] \\ &= \arg \min_{\theta} \frac{(1 - \alpha_t)^2}{2\sigma^2 \alpha_t (1 - \bar{\alpha}_t)} \left[||\epsilon_{\theta}(x_t, t) - \epsilon_0||_2^2 \right] \end{aligned}$$